

30-Second Snapshot

- Earth's relief results from a continuous balance between **endogenic uplift (aggradation)** and **exogenic weathering/erosion (gradation)**.
- Denudation** encompasses all exogenic stripping processes: in-situ **weathering**, gravity-driven **mass wasting**, kinetic **erosion**, and sediment **transportation**.
- Soil formation depends on two **active factors** (Climate, Organisms) and three **passive factors** (Parent Material, Topography, Time).
- Fluvial processes sculpt landscapes through youth, maturity, and old age, reducing highlands to a **pediplain** studded with residual **monadnocks**.
- In **karst landscapes**, calcium carbonate dissolution creates underground caves, hanging **stalactites**, rising **stalagmites**, and fused **pillars**.
- Glaciers carve **U-shaped troughs** and sharp **horns**, depositing unsorted **till moraines**, sinuous **eskers**, and streamlined **drumlins**.
- Desert landscapes reduce to **pediplains** with isolated **inselbergs**, while winds form **barchans** (horns downwind) and **parabolic dunes** (horns upwind).

Exogenic Processes & Soil-Forming Factors

Geomorphic Process / Factor	Operational Mechanism	Key Physical Signatures	Landscape / Economic Impact
Weathering (In-situ)	Mechanical fragmentation and chemical mineral breakdown without transport.	Exfoliation domes, hydration, carbonation, and corestone tors.	Prepares parent regolith for pedogenesis; concentrates bauxite ores via leaching.
Mass Movement (Gravity)	Downslope movement of regolith driven purely by gravity without a fluid medium.	Rotational slumps, planar rockslides, debris falls, and slow soil creep.	Severe hazard in the tectonic Himalayas and the rainy Western Ghats scarps.
Active Soil Controls	Climate (moisture, thermal energy) and Biological Organisms (microbes, roots).	Drives eluviation (downward leaching), illuviation (deposition), and humus creation.	Governs the chemical maturity and horizon development of the soil profile.
Passive Soil Controls	Parent Rock (texture, chemistry), Topography (slope drainage), and Time.	Thin soil on steep slopes; deep, well-differentiated horizons on flat plains.	Provides the structural mineral baseline and sets profile development limits.

Comprehensive Morphogenetic Systems Matrix

- Fluvial Geomorphology (Running Water):**
 - Erosional:** V-shaped valleys, deep gorges (canyons in horizontal strata), stream-bed potholes, plunge pools, incised meanders, and paired terraces.
 - Depositional:** Alluvial fans at mountain bases, deltas at sea mouths, active/inactive floodplains, natural levees along riverbanks, and point bars on convex banks.
 - Evolutionary Terminus:** Extensive meander loops cut through their necks to leave **oxbow lakes**, ultimately reducing the landscape to a near-flat **pediplain** with resistant **monadnock** hills.
- Karst Geomorphology (Groundwater Solution):**
 - Requires thick, jointed limestones or dolomites and abundant moisture.
 - Erosional:** Solution sinkholes, collapse dolines, coalesced uvala trenches, lapie grooves, limestone pavements, and underground caves.
 - Depositional:** Calcite precipitation creates ceiling **stalactites**, floor **stalagmites**, and fused **cave pillars**.
- Glacial Geomorphology (Moving Ice):**
 - Erosional:** Armchair-shaped cirques with tarn lakes, pyramidal horns (e.g., Matterhorn), saw-toothed arêtes, U-shaped valleys, and hanging valleys.
 - Depositional:** Unsorted glacial till forming terminal/lateral moraines, sorted glacio-fluvial outwash plains, winding esker ridges, and oval drumlin hills (steep stoss end facing up-glacier).
- Coastal Geomorphology (Marine Waves & Currents):**
 - High Rocky Coasts (India's West Coast):** Wave-cut cliffs, platforms, sea arches, and sea stacks dominate via erosion.
 - Low Sedimentary Coasts (India's East Coast):** Offshore barrier bars, spits, coastal lagoons (e.g.,

Chilika), and broad sandy beaches dominate via deposition.

• **Aeolian & Arid Geomorphology (Wind & Sheetwash):**

- **Erosional:** Sheetwash carves pediments; parallel slope retreat (backwasting) reduces ranges to isolated **inselbergs** across a flat **pediplain**; wind deflation creates deflation hollows and mushroom rocks.
- **Dune Morphologies: Barchans** (crescent horns point downwind on hard ground), **Parabolic dunes** (horns point upwind in vegetated sand), and linear **Transverse/Longitudinal** dunes.

△ **Common UPSC Exam Traps**

- **Trap 1 (Weathering vs. Erosion):** Weathering is an in-situ (on-site) process involving no transport. Erosion requires the dynamic acquisition and transportation of debris by a mobile agent (water, wind, ice).
- **Trap 2 (Mass Movement Mechanism):** Mass wasting is not driven by water or wind transporting debris. It is the downslope transfer of rock and soil under the direct pull of gravity alone, though water often acts as an internal lubricant or weight trigger.
- **Trap 3 (Soil-Forming Factors):** Climate and biological activity are **active factors** that supply moisture, heat, and organic energy. Parent material, topography, and time are **passive factors** that provide the initial mineral framework and physical conditions.
- **Trap 4 (Barchan vs. Parabolic Dune Horns):** In a barchan dune, the crescent horns point **downwind** in the direction of wind flow. In a parabolic dune, vegetation stabilizes the flanks, causing the crescent horns to point **upwind**.
- **Trap 5 (Stalactite vs. Stalagmite Growth):** Stalactites grow **downward** from cave ceilings. Stalagmites build **upward** from cave floors.

🎯 **High-Yield Facts Box**

Geomorphic Feature	Controlling Process	Diagnostic Identifier
Peneplain & Monadnocks	Fluvial erosion over time	Flat erosional plain featuring isolated resistant bedrock knolls.
Pediplain & Inselbergs	Arid sheetwash & slope retreat	Extensive desert rock plain interrupted by isolated steep inselbergs.
Drumlin	Glacial till deposition	Inverted-spoon hill; steep stoss faces up-ice, tapering tail points down-ice.
Slump	Rapid mass movement	Rotational backward slipping of cohesive debris along a curved plane.
Esker	Glacio-fluvial deposition	Sinuuous, winding ridge of stratified gravel deposited by sub-glacial streams.

📝 **Prelims Practice Challenge**

Q1. Consider the following statements regarding geomorphic processes and landforms:

1. While physical weathering involves rock disintegration through mechanical forces, chemical weathering requires the presence of water and atmospheric gases to alter minerals.
2. In a barchan dune, the two horns point upwind toward the prevailing wind source, whereas in a parabolic dune, the horns point downwind.
3. Mass movements require a mobile transport medium such as running water or glacial ice to carry weathered debris downslope.
4. Drumlins are streamlined depositional features composed of glacial till whose gentle tapering tails indicate the direction of ice movement.

Which of the statements given above are correct?

- (A) 1 and 4 only
- (B) 2 and 3 only
- (C) 1, 2, and 4 only
- (D) 1, 3, and 4 only

Answer: (A) 1 and 4 only

Explanation: Statement 1 is correct because chemical reactions require moisture and gases (O₂, CO₂) to decompose minerals, whereas physical weathering is purely mechanical. Statement 2 is incorrect because barchan horns point downwind, while parabolic horns point upwind. Statement 3 is incorrect because mass movements occur under the direct pull of gravity alone without a transporting fluid medium. Statement 4 is correct because drumlin tails taper downward in the direction of glacier flow.

Q2. With reference to pedogenesis (soil formation), which of the following are classified as active soil-forming factors?

1. Parent Material

2. Climate
3. Topography
4. Biological Organisms
5. Time

Select the correct answer using the code below:

- (A) 1, 3, and 5 only
- (B) 2 and 4 only
- (C) 2, 3, and 4 only
- (D) 1, 2, 4, and 5 only

Answer: (B) 2 and 4 only

Explanation: Climate and Biological Organisms are the two active soil-forming factors because they supply dynamic moisture, thermal energy, and organic chemical processes. Parent material, topography, and time are passive factors.

Mains Practice Question (10 Marks / 150 Words)

Question: "Differentiate between weathering, mass wasting, and erosion. Why are the Himalayas significantly more vulnerable to catastrophic landslides than the Western Ghats?"

Structured Answer Framework:

- **Introduction (25–30 words):**

Define denudation as the collective stripping of the landscape, distinguishing its three processes: weathering (in-situ rock breakdown), mass wasting (gravity-driven downslope transfer of debris without a transporting medium), and erosion (acquisition and dynamic transport of debris by water, ice, or wind).

- **Body Arguments (80–90 words):**

- *Process Distinctions:* Weathering breaks rock on-site; mass wasting moves material downslope under gravity alone; erosion uses kinetic fluid media to carve and transport sediment across distances.
- *Himalayan Landslide Vulnerability:* The Himalayas are young, tectonically active fold mountains undergoing ongoing uplift and seismic stress. They are composed largely of unconsolidated, sheared sedimentary rocks on steep slopes, which become unstable when saturated by heavy monsoon rains or disturbed by road cuts.
- *Western Ghats Structural Profile:* The Western Ghats form an ancient, stable peninsular block of hard basalt and granite. While they experience slides along steep fault scarps with deep weathering mantles during intense rainstorms, their overall frequency and structural scale are much lower than in the Himalayas.

- **Conclusion / Way Forward (25–30 words):**

Effective hill-slope management requires terrain-specific strategies: slope stabilization and strict infrastructure planning in the fragile Himalayas, combined with drainage maintenance and vegetative cover protection in the Western Ghats.